## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Measuring Research Capacity Development In Healthcare Workers: a systematic review. |
| **AUTHORS** | Bilardi, Davide; Rapa, Elizabeth; Bernays, Sarah; Lang, Trudie |

## VERSION 1 – REVIEW

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| **REVIEWER** | Giannetta, Noemi<br>University of Rome Tor Vergata |
| **REVIEW RETURNED** | 11-Dec-2020 |

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| **GENERAL COMMENTS** | Thank for the opportunity to read this review; the topic is relevant to an international audience and has considerable potential. The use of the appropriate structure frameworks for developing and presenting your work are noted and add to its value. However, there are some concerns at present that need to be addressed to enable this work to reach its potential.<br>• What does “The study was designed independently” means? Please, specify your study design in the methods section.<br>• In Prisma checklist authors state that their assessment of risk of bias of each study is presented in the main text. Please, check it, specify what tool you have used, and present the risk of bias table. |

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| **REVIEWER** | Williams, Dr Cylie<br>Peninsula Health |
| **REVIEW RETURNED** | 15-Dec-2020 |

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| **GENERAL COMMENTS** | Thank you for you work on this important topic. The authors highlight that measurement of research capacity in order to understand change and highlight need to support is important. While this review has potential, it has a clear number of deficits that I am unsure will be overcome through revision.<br><br>I would strongly urge the authors to limit the amount of acronyms that are non-standard. The paper would also benefit from a re-read for grammar and long sentences. Both issues makes the paper very hard to read as it leaves the authors continuing to try and work out what the acronym means. Long sentences also make the paper very hard to read and understand. Future revisions should spell out all of the terms.<br><br>The authors have presented the prisma checklist however there are a number of deficits within the process that have limited the believability results and how the results were obtained.<br><br>Suggestions for future editions should be:<br>- clearer identification of the research aim |

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| | - search terms not specific enough therefore missing key papers including (but not limited to):<br>Lazzarini, Peter A., et al. "Research capacity and culture in podiatry: early observations within Queensland Health." Journal of Foot and Ankle Research 6.1 (2013): 1.<br>Williams, Cylie, et al. "Research capacity and culture of the Victorian public health allied health workforce is influenced by key research support staff and location." Australian Health Review 39.3 (2015): 303-311. (referenced but not presented in table)<br>- Hulcombe, Julie, et al. "An approach to building research capacity for health practitioners in a public health environment: an organisational perspective." Australian Health Review 38.3 (2014): 252-258.<br>- Pain, Tilley, et al. "Comparison of research experience and support needs of rural versus regional allied health professionals." Australian Journal of Rural Health 23.5 (2015): 277-285.<br>- Williams, Cylie M., and Peter A. Lazzarini. "The research capacity and culture of Australian podiatrists." Journal of Foot and Ankle Research 8.1 (2015): 11.<br>- Finch, Emma, et al. "Factors influencing research engagement: research interest, confidence and experience in an Australian speech-language pathology workforce." BMC Health Services Research 13.1 (2013): 144.<br>- 16 studies were included in the results, yet only the details of 13 are in the table<br>- Authors should be urged that assessment of tool properties should be included within a systematic review. Suggest a tool such as the COSMIN, in order to describe the qualities the tool in order to best describe how the tool has been developed..<br><br>Due to the major methodological concerns, I have not reviewed the discussion in detail, as the authors should revisit the search strategy and how the included papers are reported in order to fix the current deficits. Better defining the search terms and fully exploring the literature, may result in significant changes the discussion based on the findings. |

**VERSION 1 – AUTHOR RESPONSE**

Reviewer: 1
Dr. Noemi Giannetta, University of Rome Tor Vergata, Università Vita Salute San Raffaele

Comments to the Author:
• What does “The study was designed independently” means? Please, specify your study design in the methods section.

MY RESPONSE: The sentence indicates that the PRIMA checklist on Systematic Review and CASP Critical Appraisal tool have been combined into an independently designed extraction log. To avoid misinterpretation the sentence has been removed.

• In Prisma checklist authors state that their assessment of risk of bias of each study is presented in the main text. Please, check it, specify what tool you have used, and present the risk of bias table.

MY RESPONSE: The extraction log was the principle tool used to collect and analyse the risk of

biases declared in each paper. I acknowledge the importance of making this clear to the reader and thus have revised the paragraph. Specifically, I have stated that Figure 1 only represents an extract of the extended extraction log and thus does not include the risk of biases.

Reviewer: 2
Dr. Dr Cylie Williams, Peninsula Health

Comments to the Author:

I would strongly urge the authors to limit the amount of acronyms that are non-standard.

MY RESPONSE: Thank you for this suggestion. I feel that the frequency (3 in total) of acronyms is not dissimilar to other papers published in the area of capacity development and are the same as the acronyms used in references 1, 13, 23. I think that the use of acronyms increases the readability of the paper and would therefore prefer for them to remain.

The paper would also benefit from a re-read for grammar and long sentences. Both issues make the paper very hard to read as it leaves the authors continuing to try and work out what the acronym means. Long sentences also make the paper very hard to read and understand. Future revisions should spell out all of the terms.

MY RESPONSE: All of the text has been revised and adjusted including the removal of long sentences to make the text more readable and understandable.

The authors have presented the Prisma checklist however there are a number of deficits within the process that have limited the believability results and how the results were obtained.

Suggestions for future editions should be:
- clearer identification of the research aim

MY RESPONSE: The aim of the systematic review was clearly stated in the abstract and the introduction. I acknowledge that it would have been beneficial for the aim to be stated at the beginning of the section and therefore I have moved it.

- search terms not specific enough therefore missing key papers including (but not limited to):
1. Lazzarini, Peter A., et al. "Research capacity and culture in podiatry: early observations within Queensland Health." Journal of Foot and Ankle Research 6.1 (2013): 1.
2. Williams, Cylie, et al. "Research capacity and culture of the Victorian public health allied health workforce is influenced by key research support staff and location." Australian Health Review 39.3 (2015): 303-311. MY RESPONSE: Selected but not visible in the table as per technical problem mentioned below.
3. Hulcombe, Julie, et al. "An approach to building research capacity for health practitioners in a public health environment: an organisational perspective." Australian Health Review 38.3 (2014): 252-258.
4. Pain, Tilley, et al. "Comparison of research experience and support needs of rural versus regional allied health professionals." Australian Journal of Rural Health 23.5 (2015): 277-285.
5. Williams, Cylie M., and Peter A. Lazzarini. "The research capacity and culture of Australian podiatrists." Journal of Foot and Ankle Research 8.1 (2015): 11.
6. Finch, Emma, et al. "Factors influencing research engagement: research interest, confidence and experience in an Australian speech-language pathology workforce." BMC Health Services Research 13.1 (2013): 144.

MY RESPONSE: The identification of the search strategy was a thorough, iterative process of the key and relevant words which led to the identification of 7477 relevant papers. The use of the SPIDER diagram (Table 2) and the mentioned search strategy (BOX1) provide evidence of the accuracy of this process.
I acknowledge that the papers listed by Reviewer 2 are important in this area of research. They were all identified in the initial search and were included in the 7477 selected papers, and three of them were also included in the list of 178 papers selected for full text review. However they did not meet one or more of the selection criteria (e.g. in Hulcombe et al. there is a mention of the use of a framework but the actual use of it was not described and it was not the main focus of the paper) and therefore were not included in the final set of selected papers. Williams, Cylie, et al. "Research capacity and culture of the Victorian public health allied health workforce is influenced by key research support staff and location." Australian Health Review 39.3 (2015) was selected and is included in the list of the final sixteen relevant papers and has been fully analysed in this review.

- 16 studies were included in the results, yet only the details of 13 are in the table

MY RESPONSE: Thanks you for highlighting this error which was due to a technical problem when the PDF was uploaded into the system. We will ensure the complete table is included in the re-submission.

- Authors should be urged that assessment of tool properties should be included within a systematic review. Suggest a tool such as the COSMIN, in order to describe the qualities the tool in order to best describe how the tool has been developed.

MY RESPONSE: Thank you for your suggestion of using COSMIN and we agree that this tool is useful for both the design of the review and for the evaluation of Patient-Reported Outcome Measures. However the authors decided to use the PRISMA checklist for reporting systematic reviews and meta-analysis, and the CASP checklist on critical appraisal of qualitative studies. In addition PRISMA is the most commonly used tool in submissions to Prospero. We will consider the use of COSMIN in future studies which focus more specifically on patient-reported outcome measures.

Due to the major methodological concerns, I have not reviewed the discussion in detail, as the authors should revisit the search strategy and how the included papers are reported in order to fix the current deficits. Better defining the search terms and fully exploring the literature, may result in significant changes to the discussion based on the findings.

MY RESPONSE: The importance and relevance of the work by Reviewer 2 in the field has been recognised within this systematic review and their most recent paper was included in the analysis. Unfortunately due to the technical problem of uploading the table into the system, Williams et al. (2015) was not visible in the table to Reviewer 2. I would also like to confirm this paper is also cited throughout the Results and Discussion sections.

**VERSION 2 – REVIEW**

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| **REVIEWER** | Giannetta, Noemi<br>University of Rome Tor Vergata |
| **REVIEW RETURNED** | 09-Mar-2021 |

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| **GENERAL COMMENTS** | Thank for the opportunity to read this review. The use of the appropriate structure /frameworks for developing and presenting your work are noted and add to its value. |

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| | It may be more helpful to summarize also the psychometric aspects of each tool.<br>In line with your aim, could you give some suggestions on how to adopt these instruments in practice and research? |

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| **REVIEWER** | Williams, Dr Cylie<br>Peninsula Health |
| **REVIEW RETURNED** | 15-Mar-2021 |

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| **GENERAL COMMENTS** | Thank you to the authors for being so considered in their responses and amending based on the reviewer comments. In particular correction of the missing articles and providing more detail on the inclusion/exclusion terms. This has assured me of the article inclusions and why some papers were not present in the last article<br><br>I still have a few minor points that I believe will improve the paper.<br><br>Abstract:<br>The authors have described Design as a meta-analysis. By definition, this type of analysis is a quantitative and epidemiological analysis method. I would counter you have conducted a systematic review with a narrative review. It does not take away from the robust method you have used to gather the data and synthesise in a robust way. This should be rewritten to accurately reflect the design.<br><br>While the authors have countered the use of abbreviations. I would strongly disagree on the use of non-standard abbreviations. Researchers write for clinicians and researchers and translation will be aided by easier reading. In one sentence alone, all three non-standard abbreviations were used.<br>I would highlight BMJ's suggestions of limited use of acronyms here: https://www.bmj.com/sites/default/files/attachments/resources/2018/05/BMJ-InstructionsForAuthors-2018.pdf<br>HRCD is non standard.<br>HCW is non standard and also commonly used for Healthcare Waste<br>LMIC - non standard but probably the most universally identified<br><br>Authors have intermittently described health care workers and at times health professionals. It would be helpful to align these terms.<br><br>Thank you for the additional information in the table and methods, it has helped clear my misunderstanding of in/ex criteria as to why those additional articles were not included and where they were.<br><br>Table 1 - revise for past tense<br><br>Amend postgraduates to postgraduate students.<br><br>Amend Covidence tool to Covidence systematic review software<br><br>Amend Two researchers, DB and ER independently screened the studies including or excluding studies according to the criteria in the SPIDER table (Table 1). to<br>Two researchers, DB and ER independently screened the studies, including or excluding according to the criteria in the SPIDER table (Table 1). |

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| | Remove this sentence as it is duplicated from the methods: The Covidence system presented the divergent opinions between the two researchers and disagreements were resolved with a dedicated discussion session.<br><br>Spell out US and UK the first time they are used as standard and commonly used abbreviations.<br><br>RCC and RCCT - keep these spelled out as even someone who uses these tools a lot, they get mixed around a bit.<br><br>Amend: The questionnaire was the most commonly used method for assessing research skills and competencies. to Questionnaires were the most commonly used method for assessing research skills and competencies.<br><br>I would urge reduction of conclusion as at present it is replicating much of the discussion. |

**VERSION 2 – AUTHOR RESPONSE**

Reviewer: 1
Dr. Noemi Giannetta, University of Rome Tor Vergata, Università Vita Salute San Raffaele

Comments to the Author:
Thank you for the opportunity to read this review. The use of the appropriate structure /frameworks for developing and presenting your work are noted and add to its value.

It may be more helpful to summarize also the psychometric aspects of each tool.

MY RESPONSE:
Thank you for highlighting this very interesting element for analysis. However the majority of the papers selected for inclusion did not investigate, measure or report on the psychometric aspects of each tool described. In addition, the main aim of our paper was to identify the tools available which measure health research capacity development. Therefore we feel that at this stage this aspect is beyond the scope of our paper.
We do feel it is an important point and have thus added this as a limitation to the systematic review.

In line with your aim, could you give some suggestions on how to adopt these instruments in practice and research?

MY RESPONSE:
Thank you for this comment; we have now amended the conclusion and have added a brief suggested on how such tools could be used and how each tool has to be applied taking into account different needs and circumstances.
'The applicability of each tool depends on the context and on the level of accuracy needed. Such tools could be routinely incorporated into standard personal development reviews in order to consistently support capacity development in research studies and organisations.'

Reviewer: 2
Dr. Cylie Williams, Peninsula Health

Comments to the Author:
Thank you to the authors for being so considered in their responses and amending based on the reviewer comments. In particular correction of the missing articles and providing more detail on the inclusion/exclusion terms. This has assured me of the article inclusions and why some papers were not present in the last article

I still have a few minor points that I believe will improve the paper.

Abstract:
The authors have described Design as a meta-analysis. By definition, this type of analysis is a quantitative and epidemiological analysis method. I would counter you have conducted a systematic review with a narrative review. It does not take away from the robust method you have used to gather the data and synthesise in a robust way. This should be rewritten to accurately reflect the design.

MY RESPONSE:
Thank you for suggesting the revision of the terminology; we have now amended the text to say ‘a systematic review with a narrative analysis’

While the authors have countered the use of abbreviations. I would strongly disagree on the use of non-standard abbreviations. Researchers write for clinicians and researchers and translation will be aided by easier reading. In one sentence alone, all three non-standard abbreviations were used.
I would highlight BMJ's suggestions of limited use of acronyms here:
https://www.bmj.com/sites/default/files/attachments/resources/2018/05/BMJ-InstructionsForAuthors-2018.pdf
HRCD is non standard.
HCW is non standard and also commonly used for Healthcare Waste
LMIC - non standard but probably the most universally identified

MY RESPONSE:
We completely agree that the use of abbreviations should be limited to ensure readability. Therefore we have removed all instances of ‘HCW’ and replaced these with ‘Healthcare Workers’, and all instances of ‘LMIC’ have been with ‘low and middle-income countries', throughout the text.
We hope that the reviewer would consider accepting the use of Health Research Capacity Development (HRCD) (albeit non-standard) for the following reasons:
• It is the main topic of the paper and the use of similar acronyms have been used in references 1, 13, 23.
• We attempted to replace the acronym HRCD with Health Research Capacity Development and felt it actually made the review more difficult to read and understand.
• We also attempted to reduce the concept of Health Research Capacity Development to the shorter version of "Capacity Development", but the concept of capacity development without specific reference to health research could result in misleading the reader and affect the paper's clarity.
• All of the other acronyms have now been removed.

Authors have intermittently described health care workers and at times health professionals. It would be helpful to align these terms.

MY RESPONSE:
Thank you for highlighting this point. As a result we have replaced one instance of health professionals with Healthcare Workers. In the remaining 4 cases the more generic ‘health professionals’ is required due to the context or by the specific definition (e.g. Allied Health Professionals).

Thank you for the additional information in the table and methods, it has helped clear my misunderstanding of in/ex criteria as to why those additional articles were not included and where they were.

Table 1 - revise for past tense

MY RESPONSE:
We have revised the table accordingly, thank you.

1. Amend postgraduates to postgraduate students.

2. Amend Covidence tool to Covidence systematic review software

3. Amend Two researchers, DB and ER independently screened the studies including or excluding studies according to the criteria in the SPIDER table (Table 1). to
Two researchers, DB and ER independently screened the studies, including or excluding according to the criteria in the SPIDER table (Table 1).

4. Remove this sentence as it is duplicated from the methods: The Covidence system presented the divergent opinions between the two researchers and disagreements were resolved with a dedicated discussion session.

5. Spell out US and UK the first time they are used as standard and commonly used abbreviations.

6. Amend: The questionnaire was the most commonly used method for assessing research skills and competencies. to Questionnaires were the most commonly used method for assessing research skills and competencies.

MY RESPONSE:
Thank you for these suggested revisions; we completely agree and have changed the text accordingly.

RCC and RCCT - keep these spelled out as even someone who uses these tools a lot, they get mixed around a bit.

MY RESPONSE:
We have now spelled out all of the RCC (12 instances) and the RCCT (2 instances) abbreviations throughout the text.

I would urge reduction of conclusion as at present it is replicating much of the discussion.

MY RESPONSE:
Thank you for this suggestion; we have now shortened this section to remove repetition but would still like to keep a brief summary of the findings and suggestions for future research.

**VERSION 3 – REVIEW**

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| **REVIEWER** | Giannetta, Noemi<br>University of Rome Tor Vergata |
| **REVIEW RETURNED** | 02-Apr-2021 |

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| **GENERAL COMMENTS** | Good job and good luck.<br>I accepted this article. |

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| **REVIEWER** | Williams, Dr Cylie<br>Peninsula Health |
| **REVIEW RETURNED** | 12-Apr-2021 |

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| **GENERAL COMMENTS** | Thank you for your ongoing work with this and I commend the authors for the review, it's topic and believe it will be useful to researchers who's interests closely align. |